

Tutorial 9: R Programming and Investigation on Pricing of Financial Derivatives

Theme: R Programming, AI-supported Investigation, Option Payoff, Option Value and Hedging

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Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

A. Warm-up: From Lecture Formula to R Code

Synergy & Learning Flow

Purpose of this section: The morning lecture already introduced the theory. This warm-up reconnects the mathematical formulas with R expressions. The teacher should move quickly and let students start coding as soon as possible.

Key Definition / Result

Let S_T be the future asset price and let K be the fixed price or strike price.

$$\text{Long forward payoff} = S_T - K.$$

$$\text{Call payoff} = \max(S_T - K, 0).$$

$$\text{Put payoff} = \max(K - S_T, 0).$$

In R, the function `pmax(x, 0)` calculates the element-by-element maximum between `x` and 0. This is especially useful for option payoffs when `S_T` contains many possible future prices.

Investigation / Coding Mission: Formula Translation Check

★ Core

Task. Translate the mathematical payoff formulas into R expressions.

Mathematical Formula	R Expression
$S_T - K$	<code>S_T - K</code>
$\max(S_T - K, 0)$	<code>pmax(S_T - K, 0)</code>
$\max(K - S_T, 0)$	<code>pmax(K - S_T, 0)</code>

Suggested AI Prompt: Explaining `pmax()` for Option Payoffs

★ Core

“Explain the R function `pmax()` to a secondary school student. Use the call option payoff $\max(S_T - K, 0)$ and the put option payoff $\max(K - S_T, 0)$ as examples. Please explain why ordinary `max()` is not the best choice for vectors of many possible stock prices.”

Common Mistake / Debugging Warning

In R, `max(S_T - K, 0)` returns only one single maximum value when `S_T` is a vector. For option payoff diagrams, we usually need one payoff value for each possible future price. Therefore, we use `pmax(S_T - K, 0)`.

B. Coding Mission 1: Writing Payoff Functions**Synergy & Learning Flow**

Why write functions? Instead of rewriting the same formulas again and again, students can define reusable R functions. This makes the code cleaner and helps students think like risk analysts: define the contract once, then test it under many scenarios.

Investigation / Coding Mission: Create Three Basic Payoff Functions

★

Core

Task. Write R functions for a long forward, a call option and a put option.

Teaching Note / Code Solution:

```
long_forward_payoff <- function(S_T, K) {
  S_T - K
}

call_payoff <- function(S_T, K) {
  pmax(S_T - K, 0)
}

put_payoff <- function(S_T, K) {
  pmax(K - S_T, 0)
}

# Test the functions
future_prices <- c(80, 100, 130)
```

```
K <- 100

long_forward_payoff(future_prices, K)
call_payoff(future_prices, K)
put_payoff(future_prices, K)
```

Investigation / Coding Mission: Build a Payoff Table

★ Core

Task. Use the functions to create a payoff table.

Teaching Note / Code Solution:

```
future_prices <- c(70, 80, 90, 100, 110, 120, 130)
K <- 100

payoff_table <- data.frame(
  Future_Price = future_prices,
  Long_Forward = long_forward_payoff(future_prices, K),
  Call = call_payoff(future_prices, K),
  Put = put_payoff(future_prices, K)
)

payoff_table
```

Suggested AI Prompt: Debugging Payoff Functions

★ Core

“Here is my R code for derivative payoff functions. Please check whether my formulas for long forward, call and put payoffs are correct. If there is an error, explain it simply and show a corrected version. Do not use advanced derivative pricing theory.”

Reflection Question: Function Thinking

Why is it useful to define a function before drawing graphs or running simulations?

C. Coding Mission 2: Drawing Payoff and Profit Diagrams

Synergy & Learning Flow

Visual investigation: Payoff diagrams help students see the shape of different derivative contracts. Profit diagrams then add the cost of buying the option. This is more suitable for a tutorial than repeating definitions.

Investigation / Coding Mission: Draw Forward, Call and Put Payoffs ** Intermediate

Task. Draw the payoff diagrams for a long forward, a call option and a put option.

Teaching Note / Code Solution:

```
S_T <- seq(50, 150, by = 1)
K <- 100

forward_values <- long_forward_payoff(S_T, K)
call_values <- call_payoff(S_T, K)
put_values <- put_payoff(S_T, K)

plot(
  S_T, forward_values,
  type = "l",
  col = "darkgreen",
  lwd = 3,
  ylim = c(-60, 60),
  main = "Payoff Diagrams for Three Derivative Positions",
  xlab = "Future Asset Price S_T",
  ylab = "Payoff"
)

lines(S_T, call_values, col = "blue", lwd = 3)
lines(S_T, put_values, col = "red", lwd = 3)

abline(h = 0, col = "gray40", lty = 2)
abline(v = K, col = "gray40", lty = 3)

legend(
  "topleft",
  legend = c("Long Forward", "Call Option", "Put Option"),
  col = c("darkgreen", "blue", "red"),
  lwd = 3,
  bty = "n"
)
```

Key Definition / Result

For a purchased call option:

$$\text{Call profit} = \max(S_T - K, 0) - C,$$

where C is the call premium.

For a purchased put option:

$$\text{Put profit} = \max(K - S_T, 0) - P,$$

where P is the put premium.

Investigation / Coding Mission: Write Profit Functions

★ Core

Task. Write R functions for call profit and put profit.

Teaching Note / Code Solution:

```
call_profit <- function(S_T, K, premium) {
  call_payoff(S_T, K) - premium
}

put_profit <- function(S_T, K, premium) {
  put_payoff(S_T, K) - premium
}

future_prices <- c(80, 100, 120, 140)
K <- 100
call_premium <- 8
put_premium <- 6

call_profit(future_prices, K, call_premium)
put_profit(future_prices, K, put_premium)
```

Investigation / Coding Mission: Draw Profit Diagrams

★★ Intermediate

Task. Draw call and put profit diagrams after including premiums.

Teaching Note / Code Solution:

```
S_T <- seq(50, 150, by = 1)
K <- 100
call_premium <- 8
put_premium <- 6

call_profit_values <- call_profit(S_T, K, call_premium)
```

```
put_profit_values <- put_profit(S_T, K, put_premium)

plot(
  S_T, call_profit_values,
  type = "l",
  col = "blue",
  lwd = 3,
  ylim = c(-15, 60),
  main = "Profit Diagrams for Bought Call and Bought Put",
  xlab = "Future Asset Price S_T",
  ylab = "Profit"
)

lines(S_T, put_profit_values, col = "red", lwd = 3)

abline(h = 0, col = "gray40", lty = 2)
abline(v = K, col = "gray40", lty = 3)

legend(
  "top",
  legend = c("Bought Call", "Bought Put"),
  col = c("blue", "red"),
  lwd = 3,
  bty = "n"
)
```

Common Mistake / Debugging Warning

A zero payoff does not mean zero profit. If an option expires with zero payoff, the buyer still loses the premium paid at the beginning.

Reflection Question: The Cost of Optionality

Why might someone pay a premium for a right that may not be used?

D. Coding Mission 3: Payoff, Option Value and Expected Discounted Profit

Synergy & Learning Flow

Why this section matters: Students already learned the theoretical payoff formulas in the morning lecture. In this tutorial, we use R to investigate a deeper question: What is the difference between an option payoff and an option value? The payoff is calculated at maturity after the future stock price is known. The option value today is estimated before maturity by considering many possible future payoffs, averaging them and discounting them back to today.

Important teaching choice: We still avoid formal no-arbitrage pricing, binomial trees and Black-Scholes pricing. Instead, we use a beginner-friendly scenario model. This helps students understand option value as a discounted expected payoff under stated assumptions.

Key Definition / Result

Payoff, value, premium and profit are different.

For a call option with strike K :

$$\text{Call payoff at maturity} = \max(S_T - K, 0).$$

For a put option with strike K :

$$\text{Put payoff at maturity} = \max(K - S_T, 0).$$

If possible future prices are assigned probabilities, then a simple classroom estimate of option value today is:

$$\text{Estimated option value today} = \frac{E[\text{Payoff at maturity}]}{(1 + r)^T}.$$

If the buyer pays a premium today, then:

Discounted expected profit = Estimated option value today – Premium paid today.

Therefore, if:

$$\text{Premium paid today} = \text{Estimated option value today},$$

then:

$$\text{Discounted expected profit} = 0.$$

This does not mean the actual realised profit will be zero. It only means that, under the chosen scenarios and probabilities, the expected discounted profit is zero.

Common Mistake / Debugging Warning

Do not say that option value is the same as option payoff. The payoff is a future outcome that depends on the realised future price. The value today is an estimate formed before the future price is known. It is based on possible payoffs, probabilities and discounting.

Common Mistake / Debugging Warning

Be careful with the phrase “expected value of discounted profit.” If the premium has not yet been specified, we first estimate the option value as the discounted expected payoff. Once a premium is chosen, we can calculate discounted expected profit as:

$$\text{Discounted expected payoff} - \text{premium}.$$

Investigation / Coding Mission: Scenario Table: Payoff First, Value Later

★★ Intermediate

Task. Consider three possible future stock prices. Use R to calculate the call payoff, put payoff, expected payoff and estimated option value today.

Teaching Note / Code Solution:

```
# Three possible future stock prices
S_T <- c(80, 100, 130)

# Probabilities assigned to the three scenarios
probabilities <- c(0.30, 0.40, 0.30)

# Strike price
K <- 100

# Simple annual discount rate and time to maturity
r <- 0.05
T <- 1

# Calculate payoffs in each scenario
call_payoffs <- call_payoff(S_T, K)
put_payoffs <- put_payoff(S_T, K)

# Expected payoff at maturity
expected_call_payoff <- sum(probabilities * call_payoffs)
expected_put_payoff <- sum(probabilities * put_payoffs)

# Estimated value today
```



```

call_value_today <- expected_call_payoff / (1 + r)^T
put_value_today <- expected_put_payoff / (1 + r)^T

scenario_table <- data.frame(
  Future_Price = S_T,
  Probability = probabilities,
  Call_Payoff = call_payoffs,
  Put_Payoff = put_payoffs
)

scenario_table
expected_call_payoff
expected_put_payoff
call_value_today
put_value_today

```

Reflection Question: Payoff Versus Value

In the scenario table, why is the call option payoff sometimes zero but the call option value today still positive?

Investigation / Coding Mission: Plot Premium Against Discounted Expected Profit ★★

Intermediate

Task. Draw a graph showing how the buyer's discounted expected profit changes as the premium changes.

Teaching Note / Code Solution:

```

premium_grid <- seq(0, 15, by = 0.5)

call_expected_profit_grid <- call_value_today - premium_grid

plot(
  premium_grid,
  call_expected_profit_grid,
  type = "l",
  col = "blue",
  lwd = 3,
  main = "Call Option: Premium vs Discounted Expected Profit",
  xlab = "Premium Paid Today",
  ylab = "Discounted Expected Profit"
)

```

```
abline(h = 0, col = "red", lwd = 2, lty = 2)
abline(v = call_value_today, col = "darkgreen", lwd = 2, lty = 3)

legend(
  "topright",
  legend = c("Expected Profit", "Zero Profit Line", "Estimated Option
    Value"),
  col = c("blue", "red", "darkgreen"),
  lwd = c(3, 2, 2),
  lty = c(1, 2, 3),
  bty = "n"
)
```

Suggested AI Prompt: Explaining Payoff, Value and Profit with AI ★★

Intermediate

“I am learning option pricing using simple R scenarios. Please explain the difference between option payoff, option value today, premium paid, realised profit and discounted expected profit. Use a call option example with future stock prices 80, 100 and 130, strike 100, and probabilities 0.30, 0.40 and 0.30. Avoid no-arbitrage pricing, binomial trees and Black-Scholes.”

Suggested AI Prompt: Checking My R Code for Option Value ★★

Intermediate

“Here is my R code for calculating option payoffs, expected payoff, estimated option value today and discounted expected profit. Please check whether I have confused payoff, value and profit. Explain any correction in beginner-friendly language.”

Reflection Question: Fair Premium

Under our simplified scenario model, why does paying exactly the estimated option value make the discounted expected profit equal to zero?

E. Coding Mission 4: Getting Real Stock Data in R

Synergy & Learning Flow

From invented prices to real data: Students now move from simple scenarios to real stock market data. The goal is not to make investment recommendations. The goal is to use real data as a playground for derivative investigation.

Key Definition / Result

There are two practical ways to work with real stock data:

1. **Online method:** use R packages such as `quantmod` to download market data.
2. **Offline method:** use a teacher-prepared CSV file and read it using `read.csv()`.

The online method is exciting but depends on internet access and package installation. The offline method is more reliable in classrooms.

Investigation / Coding Mission: Online Method: Download Stock Data with `quantmod`

★★ Intermediate

Task. Download Tencent stock data from Yahoo Finance and calculate daily returns.

Teaching Note / Code Solution:

```
# Install packages once if needed:
# install.packages("quantmod")
# install.packages("PerformanceAnalytics")

library(quantmod)
library(PerformanceAnalytics)

start_date <- "2021-01-01"
end_date <- Sys.Date()

getSymbols(
  "0700.HK",
  src = "yahoo",
  from = start_date,
  to = end_date
)
```

```
tencent_price <- Ad(`0700.HK`)
tencent_return <- dailyReturn(tencent_price, type = "arithmetic")

head(tencent_price)
tail(tencent_price)
summary(tencent_return)

chart.TimeSeries(
  tencent_return,
  main = "Daily Returns of Tencent",
  ylab = "Daily Return"
)
```

Investigation / Coding Mission: Offline Method: Read Teacher-prepared CSV Data

★ Core

Task. If online download does not work, read a CSV file containing date and adjusted closing price.

Expected CSV columns:

Date, AdjustedClose

Suggested AI Prompt: Debugging Real-data Import in R

★ Core

"I am trying to read stock price data into R from a CSV file. The file has columns Date and AdjustedClose. Please help me check my code, convert Date properly, calculate simple daily returns, and plot the returns. Explain each line for a beginner."

Reflection Question: Real Data

Why is real stock return data usually much noisier than the simple examples used in lecture?

F. Coding Mission 5: Historical Simulation of Future Prices

Synergy & Learning Flow

Simulation as investigation: Students now use historical returns as building blocks for possible future prices. This directly connects programming, probability and derivative pricing.

Key Definition / Result

A simple historical simulation creates future prices by resampling past daily returns:

$$S_T = S_0(1 + R_1)(1 + R_2) \cdots (1 + R_n).$$

For a one-year simulation, we often use about 252 trading days. This is a classroom convention for investigation, not a guarantee about the future.

Investigation / Coding Mission: Simulate Future Prices from Historical Returns

★★ Intermediate

Task. Simulate possible one-year future prices using resampled daily returns.

Teaching Note / Code Solution for Online Data:

```
set.seed(2026)

S0 <- as.numeric(last(tencent_price))
historical_returns <- na.omit(as.numeric(tencent_return))

number_of_simulations <- 10000
trading_days <- 252

simulated_future_prices <- rep(0, number_of_simulations)

for (i in 1:number_of_simulations) {
  sampled_returns <- sample(
    historical_returns,
    size = trading_days,
    replace = TRUE
  )

  simulated_future_prices[i] <- S0 * prod(1 + sampled_returns)
}

summary(simulated_future_prices)
```

```
quantile(simulated_future_prices, c(0.05, 0.50, 0.95))

hist(
  simulated_future_prices,
  breaks = 50,
  main = "Simulated Future Stock Prices",
  xlab = "Simulated Future Price",
  col = "lightblue",
  border = "white"
)

abline(v = S0, col = "red", lwd = 2, lty = 2)
```

Suggested AI Prompt: Explaining Historical Simulation Code

★★

Intermediate

“Explain this R code line by line. It uses historical daily stock returns, samples 252 returns with replacement, multiplies one plus the sampled returns, and creates simulated future stock prices. Please explain the financial meaning and the programming meaning separately.”

Common Mistake / Debugging Warning

Historical simulation does not say that the future will repeat the past exactly. It only uses past returns as a convenient source of possible return patterns. If future market behaviour is very different, the simulation may be misleading.

G. Coding Mission 6: Pricing Hypothetical Options from Simulated Prices

Synergy & Learning Flow

From simulated prices to simulated payoffs: Once students have many simulated future prices, option value can be investigated as a coding task: apply the payoff function to every simulated price, average the payoffs, and discount the average payoff.

This reinforces the distinction between a single realised payoff and an option value today.

Investigation / Coding Mission: Estimate At-the-money Call and Put Values

★★ Intermediate

Task. Estimate hypothetical call and put values using simulated future prices.

Teaching Note / Code Solution:

```
K <- S0
r <- 0.04
T <- 1

simulated_call_payoffs <- call_payoff(simulated_future_prices, K)
simulated_put_payoffs <- put_payoff(simulated_future_prices, K)

estimated_call_value <- mean(simulated_call_payoffs) / (1 + r)^T
estimated_put_value <- mean(simulated_put_payoffs) / (1 + r)^T

option_value_table <- data.frame(
  Option = c("Call", "Put"),
  Strike = c(K, K),
  Average_Simulated_Payoff = c(
    mean(simulated_call_payoffs),
    mean(simulated_put_payoffs)
  ),
  Estimated_Value_Today = c(
    estimated_call_value,
    estimated_put_value
  )
)

option_value_table
```

Investigation / Coding Mission: Expected Discounted Profit from Simulated Values

★★

Intermediate

Task. Suppose the call premium and put premium are chosen by the user. Calculate the discounted expected profit for the buyer.

Teaching Note / Code Solution:

```
call_premium_paid <- estimated_call_value * 0.90
put_premium_paid <- estimated_put_value * 1.10

simulated_profit_table <- data.frame(
  Option = c("Call", "Put"),
  Estimated_Value_Today = c(estimated_call_value, estimated_put_value),
  Premium_Paid = c(call_premium_paid, put_premium_paid),
```

```

Discounted_Expected_Profit = c(
  estimated_call_value - call_premium_paid,
  estimated_put_value - put_premium_paid
)
)

simulated_profit_table

```

Investigation / Coding Mission: Visualise Simulated Payoffs

★★

Intermediate

Task. Draw histograms of simulated call and put payoffs.

Teaching Note / Code Solution:

```

par(mfrow = c(1, 2))

hist(
  simulated_call_payoffs,
  breaks = 50,
  main = "Simulated Call Payoffs",
  xlab = "Call Payoff",
  col = "lightblue",
  border = "white"
)

hist(
  simulated_put_payoffs,
  breaks = 50,
  main = "Simulated Put Payoffs",
  xlab = "Put Payoff",
  col = "lightblue",
  border = "white"
)

par(mfrow = c(1, 1))

```

Investigation / Coding Mission: Mini-investigation: Compare Different Strikes

★★★ Advanced

Task. Estimate option values for different strike prices.

Teaching Note / Code Solution:

```

strike_values <- c(0.80, 0.90, 1.00, 1.10, 1.20) * S0

```



```

call_values <- rep(0, length(strike_values))
put_values <- rep(0, length(strike_values))

for (j in 1:length(strike_values)) {
  K_temp <- strike_values[j]

  call_values[j] <- mean(call_payoff(simulated_future_prices, K_temp)) /
    (1 + r)^T

  put_values[j] <- mean(put_payoff(simulated_future_prices, K_temp)) /
    (1 + r)^T
}

strike_comparison <- data.frame(
  Strike = strike_values,
  Estimated_Call_Value = call_values,
  Estimated_Put_Value = put_values
)

strike_comparison

```

Suggested AI Prompt: Interpreting Simulated Option Values

★★

Intermediate

“I estimated call and put values by applying payoff functions to simulated future stock prices and discounting the average payoff. Help me interpret the results. Explain why the values depend on the simulated price distribution and the strike price. Also remind me why this is not a professional market price.”

Reflection Question: Payoff Distribution

Why do we average the simulated option payoffs rather than average the simulated future stock prices?

H. Coding Mission 7: A First Protective-put Hedge

Synergy & Learning Flow

From pricing to hedging: Derivatives are not only things to be priced. They can also reshape risk. This section gives students a first coding investigation of a

protective put.

Key Definition / Result

A **protective put** combines:

Stock position + Put option.

For one share of stock, if the future stock price is S_T and the put strike is K_P , then:

$$\text{Put payoff} = \max(K_P - S_T, 0).$$

The hedged final value is approximately:

$$S_T + \max(K_P - S_T, 0) - \text{Put Premium}.$$

Investigation / Coding Mission: Compare Unhedged and Hedged Losses

*** Advanced

Task. Use a put with strike equal to 95% of today's stock price. Compare unhedged and hedged losses.

Teaching Note / Code Solution:

```
put_strike <- 0.95 * S0

protective_put_payoff <- put_payoff(simulated_future_prices, put_strike)

protective_put_premium <- mean(protective_put_payoff) / (1 + r)^T

unhedged_loss <- S0 - simulated_future_prices

hedged_value <- simulated_future_prices +
  protective_put_payoff -
  protective_put_premium

hedged_loss <- S0 - hedged_value

risk_comparison <- data.frame(
  Strategy = c("Unhedged Stock", "Stock with Protective Put"),
  VaR_95 = c(
    quantile(unhedged_loss, 0.95),
    quantile(hedged_loss, 0.95)
  ),
  VaR_99 = c(
    quantile(unhedged_loss, 0.99),
```

```

    quantile(hedged_loss, 0.99)
  )
)

put_strike
protective_put_premium
risk_comparison

```

Investigation / Coding Mission: Visualise Losses Before and After Hedging ***

Advanced

Task. Draw histograms of unhedged and hedged losses.

Teaching Note / Code Solution:

```

par(mfrow = c(1, 2))

hist(
  unhedged_loss,
  breaks = 50,
  main = "Unhedged Losses",
  xlab = "Loss",
  col = "lightblue",
  border = "white"
)
abline(v = quantile(unhedged_loss, 0.95), col = "red", lwd = 2, lty = 2)

hist(
  hedged_loss,
  breaks = 50,
  main = "Hedged Losses",
  xlab = "Loss",
  col = "lightblue",
  border = "white"
)
abline(v = quantile(hedged_loss, 0.95), col = "red", lwd = 2, lty = 2)

par(mfrow = c(1, 1))

```

Suggested AI Prompt: Explaining Protective Put Results *** Advanced

"I simulated future stock prices and compared unhedged stock losses with hedged losses after buying a protective put. Explain how to interpret the VaR comparison"

and the histograms. Please explain why the hedge may reduce severe losses but still has a cost."

Common Mistake / Debugging Warning

Do not conclude that the hedged position is always better just because the downside risk is lower. The premium cost, upside effect and modelling assumptions must also be considered.

Reflection Question: Risk Trade-off

If a protective put reduces severe losses but costs a premium, what kind of investor might still want to buy it?

I. Optional Extension: Two-stock Portfolio with Separate Puts

Synergy & Learning Flow

For fast groups or post-tutorial extension: This section connects the tutorial with portfolio risk. Instead of an unrealistic hypothetical put on the whole portfolio, students will simulate joint stock paths, buy separate protective puts for each stock, and visually observe the portfolio-level hedging effect.

Investigation / Coding Mission: Download a Second Stock and Align Data

*** Advanced

Task. Download HSBC data, calculate daily returns, merge it with Tencent returns, and plot them together to visually inspect their relationship before simulation.

Teaching Note / Code Solution:

```
getSymbols(
  "0005.HK",
  src = "yahoo",
  from = start_date,
  to = end_date
)

hsbc_price <- Ad(`0005.HK`)
hsbc_return <- dailyReturn(hsbc_price, type = "arithmetic")
```

```

colnames(tencent_return) <- "Tencent_Return"
colnames(hsbc_return) <- "HSBC_Return"

# Merge by date and remove NAs to keep matching days
historical_returns <- na.omit(merge(tencent_return, hsbc_return))

# Visual Aid: Plot the returns side-by-side
plot(as.zoo(historical_returns),
     main = "Historical Daily Returns: Tencent vs HSBC",
     col = c("blue", "red"))

```

Suggested AI Prompt: Merging and Plotting Financial Data*** Advanced

"I have Tencent daily returns and HSBC daily returns in R using quantmod. How do I merge them by date into one dataset, remove missing values, and create a simple base R plot showing both return series over time? Please avoid complex tidyverse functions."

Investigation / Coding Mission: Simulate Joint Future Prices

Advanced

Task. Set up a \$100,000 budget (60% Tencent, 40% HSBC). Simulate one-year future prices by resampling rows of the merged historical returns to preserve correlation.

Teaching Note / Code Solution:

```

set.seed(2026)

S0_Tencent <- as.numeric(last(tencent_price))
S0_HSBC <- as.numeric(last(hsbc_price))

budget <- 100000
shares_tencent <- (budget * 0.60) / S0_Tencent
shares_hsbc <- (budget * 0.40) / S0_HSBC

number_of_simulations <- 10000
trading_days <- 252

simulated_S_Tencent <- rep(0, number_of_simulations)
simulated_S_HSBC <- rep(0, number_of_simulations)

for (i in 1:number_of_simulations) {

```

```

# Sample row indices to keep the daily pairs together
sampled_indices <- sample(1:nrow(historical_returns), size =
  trading_days, replace = TRUE)
sampled_returns <- historical_returns[sampled_indices, ]

simulated_S_Tencent[i] <- S0_Tencent * prod(1 + sampled_returns[, 1])
simulated_S_HSBC[i] <- S0_HSBC * prod(1 + sampled_returns[, 2])
}

unhedged_portfolio_value <- shares_tencent * simulated_S_Tencent +
  shares_hsbc * simulated_S_HSBC
unhedged_loss <- budget - unhedged_portfolio_value

```

Suggested AI Prompt: Joint Historical Simulation

*** Advanced

"I want to simulate future prices for two stocks simultaneously using historical simulation in R. How can I sample the historical returns together (by rows) to preserve their correlation, and then calculate the final one-year price for each stock?"

Investigation / Coding Mission: Separate Protective Puts for the Portfolio

*** Challenge

Task. Calculate separate protective puts for both stocks at 95% of their initial prices. Find the premiums and calculate the new portfolio VaR.

Teaching Note / Code Solution:

```

K_Tencent <- 0.95 * S0_Tencent
K_HSBC <- 0.95 * S0_HSBC

# Put payoffs per share
put_payoff_Tencent <- pmax(K_Tencent - simulated_S_Tencent, 0)
put_payoff_HSBC <- pmax(K_HSBC - simulated_S_HSBC, 0)

# Premiums per share
r <- 0.04
T <- 1
put_premium_Tencent <- mean(put_payoff_Tencent) / (1 + r)^T
put_premium_HSBC <- mean(put_payoff_HSBC) / (1 + r)^T

# Total hedged portfolio value
hedged_value_Tencent <- shares_tencent * (simulated_S_Tencent +
  put_payoff_Tencent - put_premium_Tencent)
hedged_value_HSBC <- shares_hsbc * (simulated_S_HSBC + put_payoff_HSBC -
  put_premium_HSBC)

```

```

hedged_portfolio_value <- hedged_value_Tencent + hedged_value_HSBC
hedged_loss <- budget - hedged_portfolio_value

portfolio_hedge_comparison <- data.frame(
  Strategy = c("Unhedged Portfolio", "Hedged (Separate Puts)"),
  VaR_95 = c(quantile(unhedged_loss, 0.95), quantile(hedged_loss, 0.95)),
  VaR_99 = c(quantile(unhedged_loss, 0.99), quantile(hedged_loss, 0.99))
)

portfolio_hedge_comparison

```

Investigation / Coding Mission: Visualizing the Risk Reduction

Challenge

Task. Create a density plot to visually compare the distribution of losses between the unhedged portfolio and the portfolio with separate protective puts. Add vertical lines for the 95% VaR.

Teaching Note / Code Solution:

```

# Visual Aid: Compare the Loss Distributions
plot(density(unhedged_loss),
     col = "red", lwd = 2,
     main = "Portfolio Loss Distribution: Unhedged vs Hedged",
     xlab = "Loss ($)",
     xlim = c(min(hedged_loss, unhedged_loss), quantile(unhedged_loss,
0.999)))

lines(density(hedged_loss), col = "blue", lwd = 2)

# Add VaR lines
abline(v = quantile(unhedged_loss, 0.95), col = "red", lty = 2)
abline(v = quantile(hedged_loss, 0.95), col = "blue", lty = 2)

legend("topright",
     legend = c("Unhedged Loss", "Hedged Loss (Separate Puts)", "95%
VaR"),
     col = c("red", "blue", "black"),
     lty = c(1, 1, 2), lwd = 2)

```

Suggested AI Prompt: Visualizing Portfolio VaR

*** Challenge

"I have two R vectors: 'unhedged loss' and 'hedged loss' from a Monte Carlo simula-

tion. How can I use base R to plot their density curves on the same graph (red for unhedged, blue for hedged) and add vertical dotted lines showing the 95% VaR for each?”

J. AI-supported Investigation Protocol

Synergy & Learning Flow

How students should use AI in this tutorial: AI can be very helpful for explaining R code, finding syntax errors, checking whether payoff/value/profit have been confused, suggesting graph improvements and helping students interpret output. However, AI should be used as a learning partner, not as an answer machine.

Key Definition / Result

A good AI-assisted workflow:

1. Write or attempt the R code yourself.
2. Run the code and observe the output or error message.
3. Ask AI to explain the error or output.
4. Check whether the AI explanation matches the financial formula.
5. Modify the code and run it again.
6. Write a short interpretation in your own words.

Suggested AI Prompt: General Debugging Prompt

★ Core

“I am learning R for financial derivative investigation. Here is my code and the error message. Please help me identify the likely problem, explain it in beginner-friendly language, and suggest a corrected version. Please do not use advanced derivative pricing theory.”

Suggested AI Prompt: Graph Interpretation Prompt

★★ Intermediate

“I have drawn a payoff or profit diagram in R. Help me interpret the graph. Explain what happens when the future stock price is below, equal to, and above the strike price. Also explain the role of premium if the graph is a profit diagram.”

Suggested AI Prompt: Payoff, Value and Profit Checking Prompt

★★

Intermediate

“I have calculated option payoff, estimated option value today, premium paid and discounted expected profit in R. Please check whether I have confused any of these concepts. Explain the difference in simple language and point out any possible coding or interpretation mistake.”

Suggested AI Prompt: Simulation Interpretation Prompt ★★ **Intermediate**

“I used historical simulation to generate possible future stock prices and then estimated option values from average simulated payoffs. Help me explain the method, the assumptions, and the limitations in clear language for secondary school students.”

Common Mistake / Debugging Warning

Do not ask AI only for the final answer. Ask AI to explain, debug and challenge your reasoning. If AI gives a formula or code, check whether it agrees with the payoff formula from the lecture and with the distinction between payoff, value and profit.

K. Consolidation and Exit Questions**Investigation / Coding Mission: Core Coding Review**★ **Core**

1. What does $\text{pmax}(S_T - K, 0)$ calculate?
2. Why do we use $\text{pmax}()$ instead of $\text{max}()$ for payoff diagrams?
3. How do we write a reusable R function for a call payoff?
4. What is the difference between a payoff diagram and a profit diagram?
5. How does the option premium change the profit diagram?

Investigation / Coding Mission: Payoff, Value and Profit Review

★★

Intermediate

1. What is the difference between option payoff and option value today?

2. Why is option value today estimated from possible future payoffs?
3. Why is option value not simply the average future stock price?
4. What is discounted expected profit?
5. Under the simplified scenario model, why does paying the estimated option value make discounted expected profit equal to zero?
6. Why does this not mean the realised profit will definitely be zero?

Investigation / Coding Mission: Real-data and Simulation Review

★★

Intermediate

1. Why do we calculate returns from stock prices?
2. What does historical simulation do?
3. Why do we multiply many terms of the form $1 + R_i$ when simulating future prices?
4. Why may a simulated future-price histogram be useful for derivative investigation?
5. Why can the option payoff be zero in many simulated scenarios?
6. Why is the estimated option value based on average payoff rather than average stock price?

Investigation / Coding Mission: AI-supported Learning Review

★★

Intermediate

1. How can AI help when R code produces an error?
2. Why should we not blindly trust AI-generated R code?
3. What should we check after AI explains a derivative payoff formula?
4. How can AI help us interpret a graph without replacing our own thinking?
5. What is one good AI prompt you used or would use in this tutorial?

Investigation / Coding Mission: Challenge Questions

*** Challenge

1. If the strike price increases, why does the call value usually decrease?
2. If the strike price increases, why does the put value usually increase?
3. If two students choose different historical periods, why might they estimate different option values?
4. If a stock had a major crash in the historical data, how might this affect the estimated put value?
5. Why might a protective put reduce downside risk but still not be automatically the best choice?
6. How would you explain to a beginner why derivative pricing depends on assumptions?
7. Why is the phrase “fair premium” meaningful only after we state a model or set of scenarios?
8. How can AI help us understand code while still leaving the responsibility of interpretation to the student?

Key Definition / Result

Final Takeaway. The morning lecture gave the derivative theory. This tutorial turns that theory into investigation. R helps us translate formulas into functions, functions into graphs, graphs into questions, and questions into simulations. The most important conceptual distinction is this:

Payoff is a future outcome.

Option value today is a discounted expected payoff under assumptions.

Discounted expected profit is option value minus premium paid.

AI can support the learning process by explaining code, debugging errors and helping interpretation, but the student must still understand the payoff, the value, the premium, the assumptions and the financial meaning.

End of Tutorial 9